DHYEY PATEL

BACKEND ENGINEER • PYTHON & AI SYSTEMS

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PROFESSIONAL SUMMARY

Computer Science and Engineering student specializing in backend architecture and applied machine learning frameworks. Experienced in architecting scalable RESTful APIs, relational database models, and production-ready machine learning pipelines using Python, Django, FastAPI, and Flask. Demonstrated execution under pressure—validated by a 1st-place hackathon victory—with a rigorous focus on clean system design, performance engineering, and robust application security.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, SQL, HTML, CSS

Backend & Frameworks: Django, Django REST Framework (DRF), FastAPI, Flask, React, Tailwind CSS

Databases & Infrastructure: PostgreSQL, MySQL, REST API Design, Git, GitHub

Machine Learning & AI: Ensemble Methods (Random Forest, Gradient Boosting), FaceNet, KNN, Data Embeddings, OpenCV, scikit-learn, Pandas, Plotly

PROFESSIONAL EXPERIENCE

Labmentix May 2026 – Present

Python Development Intern

- Architecting and optimizing production-grade Python backend features and RESTful APIs powering enterprise AI-driven software solutions.
- Collaborating with cross-functional engineering teams to translate abstract system requirements into modular, clean, and test-driven code bases.
- Profiling application runtimes to proactively identify, isolate, and eliminate critical relational database and third-party API processing bottlenecks.

SELECTED ENGINEERING PROJECTS

EduFusionERP May 2025

Python • Django • PostgreSQL • React • Tailwind CSS

- Designed and engineered a modular enterprise resource planning architecture optimized for multi-tenant institutional academic data tracking.
- Built optimized relational schemas inside PostgreSQL, incorporating targeted indexing paradigms to minimize data query execution latencies.
- Developed a unified security layer leveraging custom middleware implementations to support strict role-based access control policies.

JalDarshi AI April 2026

Python • Flask • scikit-learn • Pandas • Plotly • JavaScript

- Architected an intelligent environmental monitoring and telemetry platform executing predictive regression analytics on live resource streams.
- Engineered high-accuracy regression models built with production-grade processing pipelines to automate data cleaning and feature isolation tasks.
- Implemented an asynchronous visualization architecture using web services to map geospatial metadata coordinates seamlessly.

Trackademic AI
April 2026

Python • FastAPI • PostgreSQL • OpenCV • FaceNet • Machine Learning

- Architected an AI-powered attendance platform utilizing FaceNet embeddings and ensemble machine learning techniques to deliver highly accurate real-time face recognition.
- Designed scalable FastAPI backend services and PostgreSQL data models to manage attendance records, user authentication, and facial embedding storage efficiently.
- Built an optimized computer vision pipeline using OpenCV for face detection, preprocessing, and feature extraction, enabling low-latency recognition suitable for production deployment.

Procura
December 2024

Python • Flask • MySQL • React • REST API

- Developed a centralized commercial inventory routing backend engine engineered to track logistical supply chain changes deterministically.
- Authored transactional integrity safeguards in MySQL to prevent race conditions during highly concurrent resource state modifications.
- Constructed clean RESTful API endpoint designs following strict OpenAPI structures to enable fluid client-side model synchronization.

Signly
March 2025

Python • OpenCV • scikit-learn • NumPy • Pandas

- Engineered an isolated computer vision translation pipeline to transform matrix-represented visual gestures into clean textual character arrays.
- Designed optimized input image processing frameworks via mathematical color masking and threshold filters to reduce feature matrix spaces.
- Refactored machine learning model evaluation workflows to evaluate streaming data frames efficiently with minimal frame drop counts.

EDUCATION

Karnavati University
Expected Graduation: May 2028

B.Tech in Computer Science and Engineering

- Core Focus: Advanced Data Structures & Algorithms, Backend Architectural Paradigms, Systems Engineering.

HONORS & AWARDS

- **1st Place Winner** — Trackademic AI Hackathon (April 2026)
- **Official Selection** — HackOdisha 5.0 Core Infrastructure Hackathon Finalist (2025)